Assignment Solution Explanation

1. The Student Class (Student.java)

The Student class serves as the blueprint for student objects, focusing heavily on encapsulation and proper method overriding:

- Encapsulation: All fields (name, age, grades) are declared as private to protect the data. Access is provided through public getters and setters.
- Constructors: I implemented three constructors, including a default constructor (setting default values like "Unknown" or null) and a parameterized constructor to allow flexible object creation.
- Overriding Methods:
    - **toString**(): Overridden to return a readable string format of the student's data using Arrays.toString().
    - **equals**(): Logic was implemented to compare the actual values of the fields rather than memory addresses.
    - **hashCode**(): I used Objects.hash() for the name and age, then manually incorporated Arrays.hashCode(grades) multiplied by 31 (searched and found out that 31 is commonly used in hashing) to ensure the array contents are part of the unique hash.
- Internal Logic: The class includes a **calculateAverage**() method using the **this** keyword to provide the average grade for an individual student.

---

2. The Service Layer (StudentService.java)

The **StudentService** class contains the business logic for managing a group of students:

- Student Creation: The **createStudent** method acts as a factory to instantiate new Student objects.

- Analytics: I implemented **getTopStudent** to find the student with the highest average and **getClassAverage** to calculate the mean score of the entire array.
- Method Overloading: As required, the **calculateAverage** method is overloaded. One version accepts a raw int[] array, while the other accepts a Student object.

---

3. Application Execution (Main.java)

The Main class serves as the testing environment for the system:

- Data Storage: I created an array to store three Student objects as the minimal requirement.
- Processing: A **for-each** loop is used to iterate through the student array and print their details and averages to the console.
- Results: The program concludes by displaying the calculated class average (formatted to two decimal places) and identifying the top-performing student.